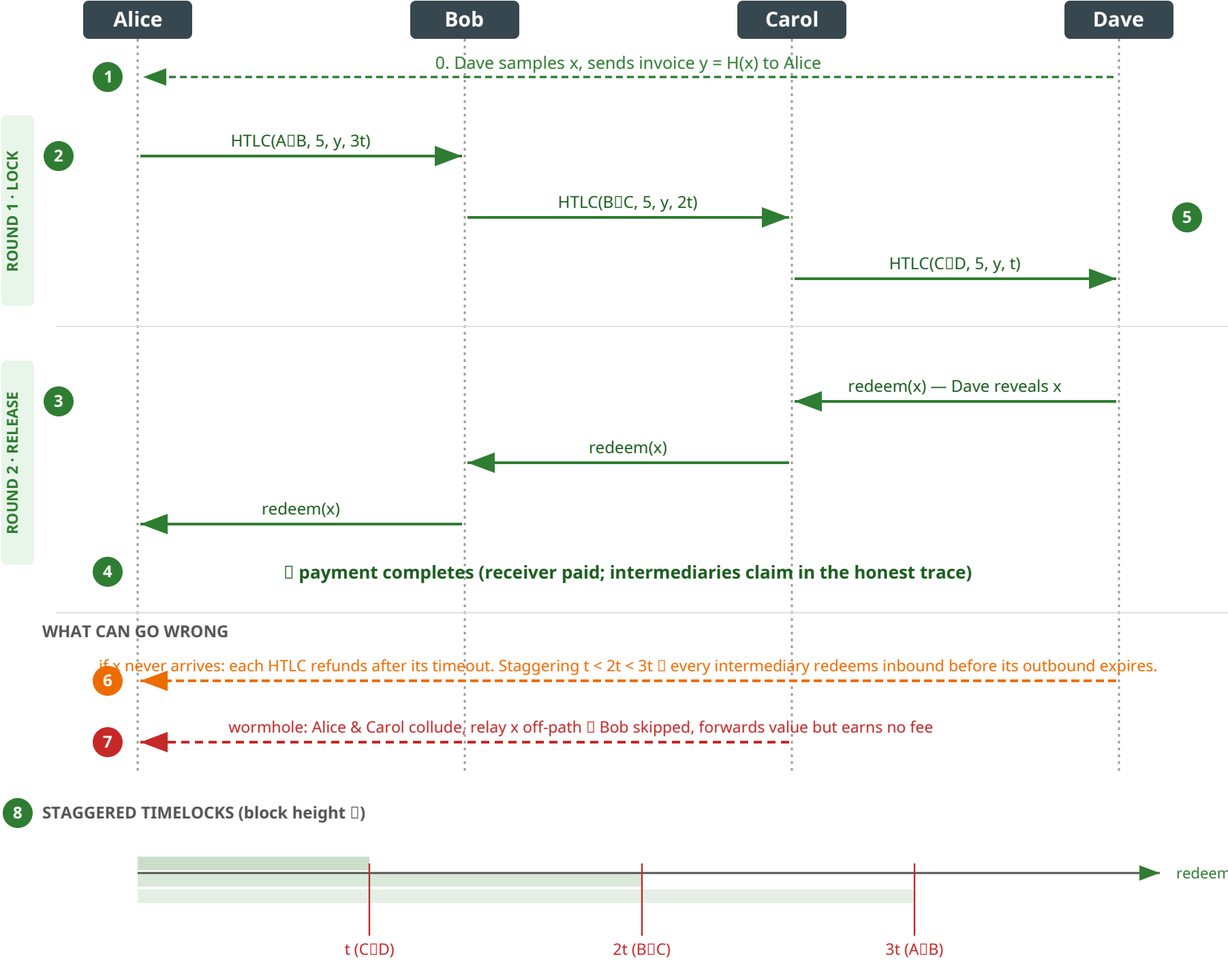




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1 Invoice & preimage authentication

Invoice_Authenticates_Settlement · Preimage_Secret_Until_Release · Forged_Invoice_Requires_Key_Compromise · Invoice_Released_On...

2 Lock causality (forward needs previous)

Forward1_Requires_Offer · Forward2_Requires_Forward1 · HTLC_On_Opened_Channel $\square$

3 Release causality (redeem needs preimage)

Fulfill_Requires_Forward2 · Claim_Requires_Release · Settle_Requires_Receiver_Release $\square$

4 Atomicity + non-vacuity

Payment_Atomicity_Under_Liveness $\square$ · Multihop_Payment_Possible $\square$ (witness)

5 Value & fee conservation

EndToEnd_Value_Conservation · Fee_Conservation_Hop1/2 · Fee_Strictly_Positive_Hop1/2 · Receiver_Paid_Invoice_Amount $\square$

6 Refund / grieving

Refund_Requires_Timeout · Loss_Requires_Inaction $\square$ · Griefing_Capital_Lock_Reachable $\square$ (reachable DoS)

7 Wormhole fee theft

Wormhole_Fee_Theft_Reachable $\square$ · Wormhole_Steals_Exactly_The_Fees $\square$ (theft = path fees)

8 Intermediary safety (under liveness T2b)

Intermediary_Never_Loses_Under_Liveness $\square$ · T2b_Counterexample_Blocked $\square$ (race blocked with T2b)

$\square$ = machine-checked in multihop.spthy (43/43).
green = all-traces safety · orange = benign/reachable
red = attack shown reachable (by design, no PTLC).

Channel-layer lemmas (open/update/close/punish)
live on the Alice↔Bob figure.